Module 1 — Digital Forensics (Exam-Oriented Notes)

Updated as per teacher's official PPT (GLA University — Priyanka Sharma)

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1. Foundations of Digital Forensics

Definition

Digital Forensics is the process of **uncovering, preserving, acquiring, analysing, and presenting** digital evidence in a legally admissible manner. It applies scientific techniques to digital devices to reconstruct past events during an investigation.

> **Formula (exam favorite):** `Science/Technology + Legal Standards = Admissible Evidence`

Core Objectives (guaranteed 2-5 mark question)

- **Preserve Evidence Integrity** — evidence unaltered from seizure to court
- **Identify Cause and Intent** — who, what, when, where, how
- **Reconstruct Timelines** — chronological sequence of events
- **Maintain Chain of Custody** — document every individual who handles evidence
- **Ensure Legal Admissibility** — evidence must withstand courtroom scrutiny

Digital Evidence — Definition

Digital evidence (electronic evidence) is any **binary data or information** stored, transmitted, or processed by an electronic device that can be relied upon in court or an internal investigation to prove/disprove a fact.

- **ISO/IEC 27037** standard: digital evidence must maintain its **probative value** (ability to prove a point) through rigorous acquisition, handling, and analysis practices.

Locard's Exchange Principle (IMPORTANT — direct question aata hai)

> *"Anyone or anything, entering a crime scene takes something of the scene, and leaves something of themselves behind."*

Yeh principle digital forensics ka foundation hai — koi bhi digital interaction (login, file access, USB connect) apna trace chhod jaata hai (logs, timestamps, metadata).

Key Characteristics of Digital Evidence (compare with physical evidence — likely question)

Unlike physical evidence (fingerprints, blood samples), digital evidence has unique properties:

Property	Meaning	Example
Fragile & Volatile	Easily altered/overwritten/destroyed	Booting a computer changes system timestamps and overwrites RAM
Latent (Hidden)	Not visible to naked eye	Needs software to extract metadata, deleted files, unallocated space
Easily Duplicable	Exact 1:1 copies (bit-stream images) possible without degrading original	Forensic imaging
Time-Sensitive	Volatile memory/network sessions vanish when power cut or overwritten	RAM data lost on shutdown

> **Example flow:** Employee copies confidential files to USB → `USB device → File → Metadata → Timestamp → User activity → Evidence`

Types of Digital Evidence

Type	Description	Examples
Volatile Data	Temporary info, needs constant power, deleted on power loss	System time, logged-on users, open files, network info, process info, process-to-port mapping, clipboard contents, command history
Non-volatile Data	Permanent data on secondary storage	Hidden files, slack space, swap file, index.dat files, unallocated clusters, unused partitions, registry settings, event logs

Major Types of Digital Forensics (8 types — table question expect karo)

Type	Focus
Disk Forensics	Hard drives, storage media
Network Forensics	Traffic, packets, logs
Database Forensics	Database records, transactions
Memory Forensics	RAM analysis
Mobile Forensics	Smartphones, SIM, apps
Malware Forensics	Malicious code analysis
Email Forensics	Email headers, spoofing, phishing trace
Cloud Forensics	Cloud-stored data, logs, accounts

(All 8 fall under the umbrella of "Computer Forensics")

The 5-Phase Forensic Process (VERY IMPORTANT — 10-mark question guaranteed)

```
1. Identification → 2. Collection → 3. Preservation → 4. Analysis → 5. Reporting
```

> **Golden Rule:** Always work on a forensic copy — the **original evidence is never modified.**

1. Identification

- Recognize potential evidence sources: computers, servers, mobile devices, cloud accounts, IoT devices, removable media
- Determine scope — what data is relevant
- Assess **volatility** — RAM/network sessions disappear if powered off, so prioritize them
- Secure the scene to prevent alteration before collection

2. Collection

- Acquire data using forensically sound methods — **bit-by-bit (bitstream) image**, NOT a simple file copy
- Obtain legal authorization first (warrant, consent, exigent circumstance)
- Use **write-blockers** to prevent write operations to original media
- Start chain-of-custody log the moment evidence is touched

3. Preservation

- Protect evidence from alteration/damage/deletion for the case duration
- Generate **cryptographic hash values** (MD5, SHA-1, SHA-256) immediately after acquisition — this is the "fingerprint"
- Store original + verified image in secure, access-controlled storage
- Re-verify hashes at each access stage

4. Analysis

- Work only on the **verified copy/image** — original never touched again
- Recover deleted files, examine file system metadata, registry, logs, browser/app artifacts
- Reconstruct timelines by correlating timestamps
- Use validated tools (EnCase, FTK, Autopsy) so methodology withstands scrutiny

5. Reporting

- Clear, structured, objective report — methodology, tools, evidence handled, conclusions
- Write for two audiences: technically precise (peer review) + understandable (judges/juries/executives)
- Include chain-of-custody trail + hash verifications
- Be prepared to defend findings under cross-examination

Digital Evidence Lifecycle (7-step version — different from 5-phase process, dono yaad rakhna)

```
Identification → Collection → Preservation → Examination → Analysis →
Documentation → Presentation
```

> **Exam trap:** Yeh 7-step "lifecycle" hai vs 5-phase "forensic process" — dono naam alag context me diye jaate hain, agar question specifically "lifecycle" bole to yeh 7-step wala likhna.

Checklist to Prepare for a Computer Forensics Investigation

- Do NOT turn the computer off/on, run programs, or access data before securing it
- Secure all relevant media (hard drives, phones, DVDs, USB drives)
- Suspend document destruction/recycling policies for relevant media/users
- Perform preliminary crime-scene assessment — identify data type, urgency level
- Once machine is secured, obtain info about machine, peripherals, network
- If possible, obtain passwords for encrypted/password-protected files
- Compile list of names, emails, contacts the subject communicated with

- If computer was accessed before securing, note who accessed it, what files, when, why
- Maintain chain of custody for every piece of original media
- Create a list of keywords/phrases for searching relevant data

Chain of Custody (HIGH-WEIGHTAGE — almost guaranteed question)

- **Legal document** that demonstrates the progression of evidence from original location to forensic lab
- Roadmap showing how investigators collected, analyzed, and preserved evidence
- Ensures accurate auditing of original data, imaging of source media, log tracking

Legal basis (India) — CORRECTED:

> **Section 63 of the Bharatiya Sakshya Adhiniyam, 2023 (BSA)** governs admissibility of electronic records in Indian courts — this has **replaced the old Section 65B of the Indian Evidence Act**. It permits computer/digital output (emails, WhatsApp chats, CCTV footage) as evidence **without producing the original device**, provided a mandatory certificate accompanies it.

Forms used (may be asked to identify/draw):

- **Single Evidence Form** — Case No., Evidence No., Date/Time Collected, Collected by, Site Address, Storage Location, Device Type, Manufacturer, Model, Serial No., **MD5 Sum, SHA-1 Sum**, Image Details
- **Chain of Custody Form** — accompanies Single Evidence Form; tracks Submitter → Receiver with Name, Signature, Date & Time, and "Evidence Modified: Yes/No" for every handoff

Role of Investigators

- Evaluates damages of a security breach
- Identifies and recovers data required for investigation
- Extracts evidence in a forensically sound manner
- Ensures proper handling of evidence
- Acts as a guide to the investigation team
- Creates reports/documents for court presentation
- Reconstructs damaged storage devices, uncovers hidden information
- Updates organization about attack methods and recovery techniques (maintains records)
- Testifies in court, defends findings

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2. Cyber Laws (COMPLETE SECTION — was missing earlier, high weightage)

Definition & Objectives

Cyber laws are legal rules governing computers, digital devices, networks, internet, and electronic communications — designed to prevent cybercrimes, protect digital information, safeguard privacy, regulate e-transactions, and ensure secure IT use.

Objectives:

- Protect systems/networks/data from unauthorized access
- Legal recognition to electronic records and digital signatures
- Prevent and punish cybercrimes
- Ensure privacy and security
- Facilitate secure e-commerce/online banking
- Establish investigation/prosecution procedures
- Promote ethical IT use

Information Technology (IT) Act, 2000 (India's primary cyber law)

Came into force **17 October 2000**; amended in 2008.

Important Sections (ratta maar — table question likely):

Section	Deals With
43	Compensation for unauthorized access, virus introduction, damaging systems
65	Tampering with computer source code
66	Hacking / unauthorized access (dishonest/fraudulent)
66C	Identity theft (passwords, digital signatures)
66D	Cheating by personation (online fraud, phishing)
66E	Privacy violation — capturing/publishing private images without consent
67	Publishing/transmitting obscene material electronically
69	Government's power to intercept, monitor, decrypt information
72	Breach of confidentiality and privacy

IT (Amendment) Act, 2008 (ITAA 2008)

Change	Section(s)	Impact
Electronic Signatures	3A	Additional secure authentication methods beyond digital signatures
Electronic Contracts	10A	Legal recognition to contracts via email/website
Data Protection	43A	Compensation liability for negligent data breach
Privacy Protection	72A	Punishes unauthorized disclosure of personal info
New Cyber Crimes	66B–66F, 67A, 67B	Identity theft, cyber terrorism, privacy violation, CSAM
CERT-In	70B	National agency for cybersecurity incident response
Safe Harbour	79	Legal immunity for intermediaries (ISPs, platforms) with due diligence
Surveillance & Blocking	69A, 69B	Govt power to intercept/block content
Critical Infra Protection	70A	Protects banking, power, telecom, defence sectors

International Laws (comparison table — exam favorite)

Law	Region/Year	Key Focus
GDPR	EU, 2018	Personal data protection; user consent; "Right to be Forgotten"; 72-hour breach reporting; heavy penalties
HIPAA	USA, 1996	Protects healthcare/medical data (PHI); Privacy/Security/Breach Notification Rules
CCPA	California USA, 2020	Consumer right to know/delete data; opt-out of data sale
Budapest Convention	Council of Europe, 2001	First international cybercrime treaty; harmonizes laws; supports cross-border evidence sharing

> **Exam tip:** GDPR "Right to be Forgotten" aur "72-hour breach reporting" — yeh 2 points bahut common questions me aate hain.

3. Evidence Collection & Preservation

Evidence Collection — Definition

Obtaining digital information from a device while maintaining its **original state and integrity**.

Potential evidence sources: Computer, Laptop, Mobile phone, HDD/SSD, USB, SD card, Cloud account, Network device, Server, IoT device

Evidence Collection Methods

A) Live vs Dead Forensics

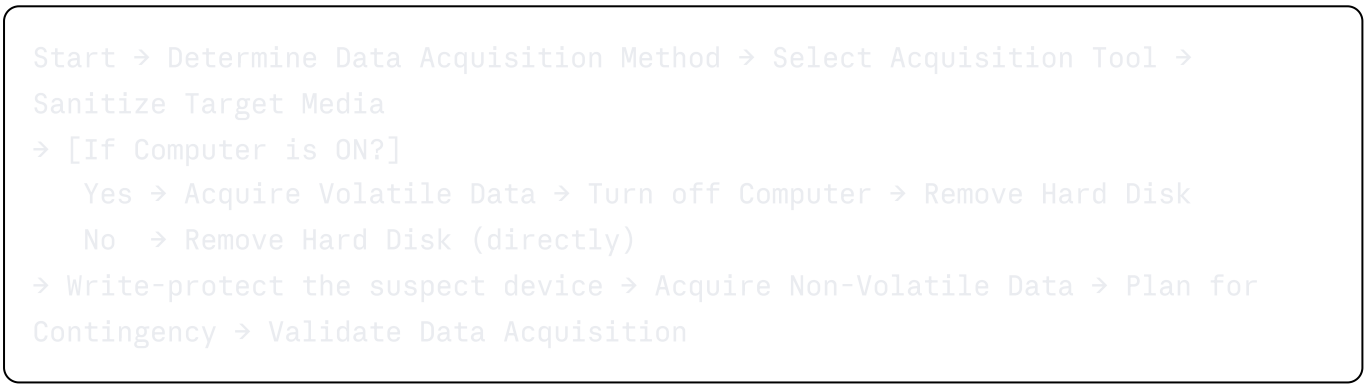
	Live Acquisition	Dead/Offline Acquisition
When	System is running	System powered off
Collects	RAM, running processes, network connections, encryption keys, active sessions	Storage media (via write blocker)
Advantage	Captures volatile data	Minimizes modification, easier integrity preservation

> **Process (Dead acquisition):** Computer → Remove HDD → Connect using Write Blocker → Create forensic image

B) By Scope

- **Physical Acquisition** — bit-by-bit copy of entire storage device
- **Logical Acquisition** — selected files/folders/user data only
- **Network Acquisition** — network packets/logs (e.g., PCAP, firewall logs)
- **Cloud Acquisition** — emails, cloud files, login records

Evidence Collection Flowchart (may be asked to draw/explain)



Write Blocker

A hardware/software mechanism that **prevents data from being written** to the original evidence storage device.

Without it, forensic workstation could unintentionally modify:

- File timestamps
- File system metadata
- System files
- Access information

Duplication of Data — Imaging vs Cloning (comparison — common question)

Disk Imaging	Disk Cloning
Creates an **image file** of the disk (e.g., `.img`, `.E01`)	Creates a **direct copy** onto another disk (HDD → HDD)
Can be stored on computer/server	Requires another physical storage device
Commonly used in digital forensics	Commonly used for disk duplication/deployment
Original evidence remains separate	Target becomes a copy of the source
Easy to create multiple working copies from one image	Usually creates one direct disk copy

Integrity Verification

Process of ensuring digital evidence has **not been altered, modified, or corrupted** during acquisition, examination, or storage.

Why important:

- Maintains authenticity
- Detects accidental/intentional modification
- Ensures court evidence = originally collected evidence
- Supports chain of custody
- Establishes forensic reliability

Basic principle:

```
Original Evidence → Hash Calculation → Examination → Hash Recalculation → Comparison
```

- Hashes match → evidence unchanged ✅
- Hashes differ → evidence modified/corrupted ❌

Hashing (GUARANTEED QUESTION)

A hash function converts data of any size into a **fixed-length value** (hash value/digest).

Algorithm	Hash Length	Current Forensic Use
MD5	128 bits / 32 hex characters	Legacy; compatibility use, not ideal as sole integrity proof
SHA-1	160 bits / 40 hex characters	Legacy; collision attacks make it unsuitable as primary modern algorithm
SHA-256	256 bits / 64 hex characters	**Preferred modern choice**

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4. System & Storage Analysis

Understanding Hard Disk Drive (HDD)

- Non-volatile storage; records data **magnetically** on metallic platters
- Read/write performance directly proportional to **RPM** (revolutions per minute)
- Has moving parts → susceptible to physical damage/wear over time

HDD Parts (diagram-based question possible):

Part	Description
Platter	Circular metal disk storing data
Track	Concentric ring on platter; numbering starts at 0
Sector	Smallest physical storage unit, **512 bytes**; new format = 8×512B → one 4096B (4KB) sector
Cluster	Smallest logical/accessible storage unit; combines sectors (2–32+ depending on formatting)
Cylinder	Group of tracks at same position across all platter surfaces

CHS (Cylinder-Head-Sector) Addressing & Capacity Calculation (NUMERICAL — likely to appear!)

Formula:

```
Total Disk Size = No. of Cylinders × No. of Heads × Sectors per Track × Bytes per Sector
```

> **Solved Example (exactly as in PPT — practice this):** Disk has 16,384 cylinders, 80 heads, 63 sectors/track, 512 bytes/sector. Total Size = 16,384 × 80 × 63 × 512 = **42,278,584,320 bytes**

Understanding Solid-State Drive (SSD)

- Non-volatile; uses **NAND flash memory chips**
- Faster than HDD — no moving parts

SSD Parts:

Part	Function
NAND Flash Memory	Main storage — floating gate transistors retain charge without power
Controller	Processor bridging flash memory and host (firmware-level)
DRAM	Volatile memory for faster read/write
Host Interface	Connection type — SATA, PCIe, SCSI

> **HDD vs SSD one-liner for exam:** HDD = mechanical (platter/spindle/actuator arm), SSD = electronic (flash memory/controller/cache) — SSD is faster, no moving parts, but HDD is generally cheaper per GB.

Lost Clusters

- When OS marks clusters as used but doesn't allocate them to any file
- **A FAT file system error** — result of logical structure error, NOT physical disk error
- Occurs due to interrupted file activities (file not properly closed)
- **CHKDSK** — Windows system tool that authenticates file system reliability and repairs logical errors

Slack Space (NUMERICAL EXAMPLE — important)

Storage area between **end of file** and **end of cluster**. If file size < cluster size, full cluster is still allocated — remaining space = slack space.

> **Example:** Partition size = 4GB → cluster size = 32KB. File needs only 10KB → entire 32KB allocated → **22KB slack space** created (old/deleted data may hide here — evidence source!)

Master Boot Record (MBR)

- First sector ("**sector zero**") of a storage device
- Stores info about files, locations, sizes on disk
- Usually refers to the **512-byte boot sector**
- Important for both booting AND forensics

Forensic importance — MBR analysis reveals:

- Partition information & disk layout

- Boot configuration
- Evidence of boot-sector modification
- Possible bootkits/boot-sector malware
- Deleted/altered partition information
- Original disk structure

Globally Unique Identifier (GUID)

- **128-bit value** uniquely identifying an object, device, partition, file, or component
- Represented as **32-character hexadecimal** value in 5 groups
- Important with **GPT (GUID Partition Table)** — each GPT partition has a unique Partition GUID, identifiable independent of physical location

File Carving

Technique to recover files/fragments **in the absence of file system metadata** — searches raw data for file signatures (headers/footers).

- **File header** = signature (constant numeric/text value) determining file format
- Tools to verify format: 010 Editor, Hex Viewer, WinHex

Useful when:

- File deleted
- File system corrupted
- Metadata missing
- Device formatted
- Examining unallocated space
- File fragments remain even after directory entry gone

Importance in Digital Forensics:

- Recovers deleted files from unallocated space
- Works without MFT/directory entries
- Recovers evidence after formatting/corruption
- Identifies hidden/renamed files via signatures
- Extracts images, videos, PDFs, documents
- Useful for damaged devices/forensic images
- Helps reconstruct digital activities

File Systems — Detailed Comparison (HEAVY WEIGHTAGE)

FAT Family:

Version	Introduced	Max Partition Size	Max File Size	Filename
FAT12	Early 1980s	~16 MB	~16 MB	8.3 format
FAT16	—	~2 GB	~2 GB	8.3 format
FAT32	Windows 95 OSR2	~2 TB (Windows)	4 GB − 1 byte	Up to 255 chars (LFN)
exFAT	Microsoft, 2006	~128 PB	~16 EB − 1 byte	Up to 255 chars

> **Exam tip:** FAT32's 4GB file-size limit is a very commonly tested number — memorize it.

NTFS (New Technology File System) — default for modern Windows:

- **Boot Sector** — cluster size, MFT location info
- **MFT (Master File Table)** — main database; stores file/directory records & metadata
- **MFT Zone** — reserved space for MFT growth
- **System Files** — `$LogFile`, `$Bitmap`, `$Boot`, `$UsnJrnl`
- **Data Area** — actual file contents in clusters
- Volume size: up to ~8 PB; Default cluster size: 4KB (up to 64KB); Filename: 255 chars, Unicode supported

FAT vs NTFS (comparison table — very likely exam question):

Point	FAT	NTFS
Structure	File Allocation Table	MFT (Master File Table)
File Size	FAT32: max 4GB−1byte	Supports very large files
Partition Size	FAT32: ~2TB	Much larger volumes
Security	❌ No advanced permissions	✅ ACL-based security
Journaling	❌ Not supported	✅ Supported
Reliability	Less reliable after crashes	More reliable (journaling)
Encryption	❌ No native EFS	✅ EFS supported
Compression	❌ Not supported	✅ Supported
Forensics	Less metadata	Rich forensic metadata

Apple File Systems:

System	Introduced	Key Points
HFS	1985	Legacy; hierarchical structure; max file size ~2GB; historical forensics only
HFS+	1998	Improved HFS; Unicode filenames; journaling support; max file size ~8EB (theoretical); largely replaced by APFS
APFS	2016 (macOS High Sierra 2017)	Modern; optimized for SSD/flash; used across macOS/iOS/iPadOS/watchOS; supports **Copy-on-Write, Snapshots, Clones, Space Sharing, strong encryption, checksums**

Windows Forensics — 4 Key Artifacts (intro — detailed in next unit)

- **Windows Registry** — system/user config database
- **Windows Event Logs** — login attempts, system/security events
- **Windows Prefetch** — program execution history
- **Windows Shortcuts (LNK files)** — recently accessed file/folder trace

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5. Data Recovery & Storage Forensics

Deleted File Recovery

Deleting a file only removes its **reference in the file table** (e.g., MFT entry in NTFS) — actual data stays on disk until **overwritten** by new data.

Encrypted & Hidden Data Retrieval

- **Encrypted data** — BitLocker, VeraCrypt volumes; examiner needs key/password or brute-force/dictionary attack
- **Hidden data** — Steganography (data hidden inside image/audio), NTFS Alternate Data Streams (ADS)

> **Example:** Suspect hides text inside an image via steganography — invisible on normal viewing, extractable via tools like steghide, binwalk.

Recovery Tools

- **FTK Imager** — imaging + preview of deleted files
- **EnCase** — industry-standard, comprehensive suite (imaging + analysis + reporting)

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6. Malware & Threat Analysis

Malware Types

Type	Behavior
Virus	Self-replicates by attaching to files
Worm	Self-replicates independently, spreads via network
Trojan	Disguised as legitimate software, gives backdoor access
Ransomware	Encrypts files, demands payment
Spyware	Silently monitors user activity

Static vs Dynamic Analysis

	Static Analysis	Dynamic Analysis
Method	No code execution (disassembly, string analysis)	Actual execution in controlled environment
Risk	Safe	Risky — needs isolated sandbox
Info gained	Code structure, strings, signatures	Runtime behavior — file changes, network calls, registry edits

Sandboxing (Cuckoo Sandbox)

Isolated virtual environment for safely executing malware to observe behavior (files created, network connections) without harming real systems.

Reverse Engineering Basics

Disassembling/decompiling malware binaries to understand internal logic.

- **IDA Pro** — disassembler, converts machine code to assembly
- **Ghidra** — NSA's free open-source RE tool

Process Injection & Hidden Threats

Malware injects malicious code into a legitimate running process (e.g., `explorer.exe`) to:

- Avoid antivirus detection
- Use the legitimate process's privileges

> **Example:** DLL Injection — malicious DLL loaded into a trusted process's address space, so security tools see it as "trusted."

Quick Revision Table (last-minute cramming)

Topic	Must-remember keyword
Definition formula	Science + Legal Standards = Admissible Evidence
Locard's Principle	Every interaction leaves a trace
Digital evidence properties	Fragile, Latent, Duplicable, Time-Sensitive
5-Phase Process	Identification → Collection → Preservation → Analysis → Reporting
7-Step Lifecycle	Identification → Collection → Preservation → **Examination** → Analysis → **Documentation** → Presentation
Legal basis (India)	**Section 63 BSA 2023** (replaced Section 65B IEA)
IT Act key sections	43, 65, 66, 66C, 66D, 66E, 67, 69, 72
International laws	GDPR (EU), HIPAA (health, USA), CCPA (California), Budapest Convention
Hashing	MD5 (128-bit) → SHA-1 (160-bit) → SHA-256 (256-bit, preferred)
CHS Formula	Cylinders × Heads × Sectors/Track × Bytes/Sector
Slack space example	4GB partition, 32KB cluster, 10KB file → 22KB slack
FAT32 limit	Max file size = 4GB − 1 byte
FAT vs NTFS	NTFS has journaling, ACL security, EFS, rich metadata
Apple FS evolution	HFS → HFS+ → APFS (SSD-optimized)
Imaging vs Cloning	Imaging = image file; Cloning = disk-to-disk copy
Recovery logic	Delete = pointer removed, data stays till overwritten
Malware analysis	Static (no execution) vs Dynamic (sandbox execution)
RE tools	IDA Pro, Ghidra

Pro tip for exam: Numerical questions (CHS capacity, slack space calculation) are **easy scoring** if formula is memorized — practice 2-3 variations. For theory questions, always mention the **exact section number** (Section 63 BSA, not old 65B) — professors specifically check for this update.